

A Scalable Multimodal Big Data Framework for Transformer-Based Emotion Recognition

Applications in Mental Health Analytics for High-Stress Environments



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OUTLINE

1. INTRODUCTION

2. RELATED WORK

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5. CONCLUSION

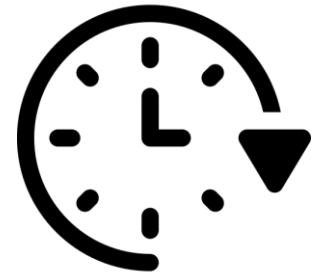


The Imperative: Mental Health in the Data Era

- High-stress professionals (e.g., emergency care) face critical risks of burnout and depression.
- Current monitoring is subjective, intermittent, and lacks real-time precision.
- WHO Global Context: 280+ million people suffer from depression (3.8% of the global population).
- The Gap: Organizations lack frameworks to transform digital footprints into actionable health indicators.



Stress



Long Hours

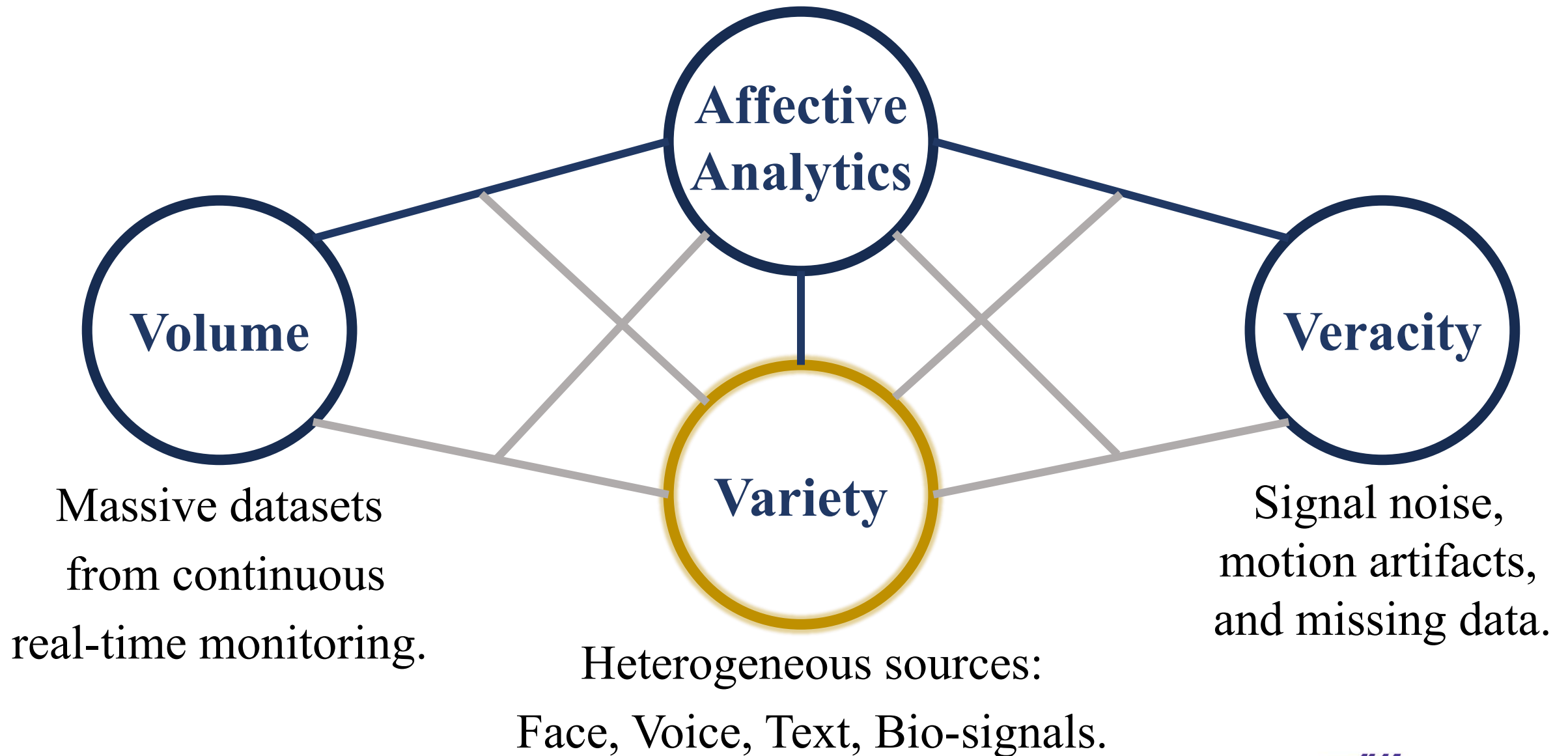


Burnout

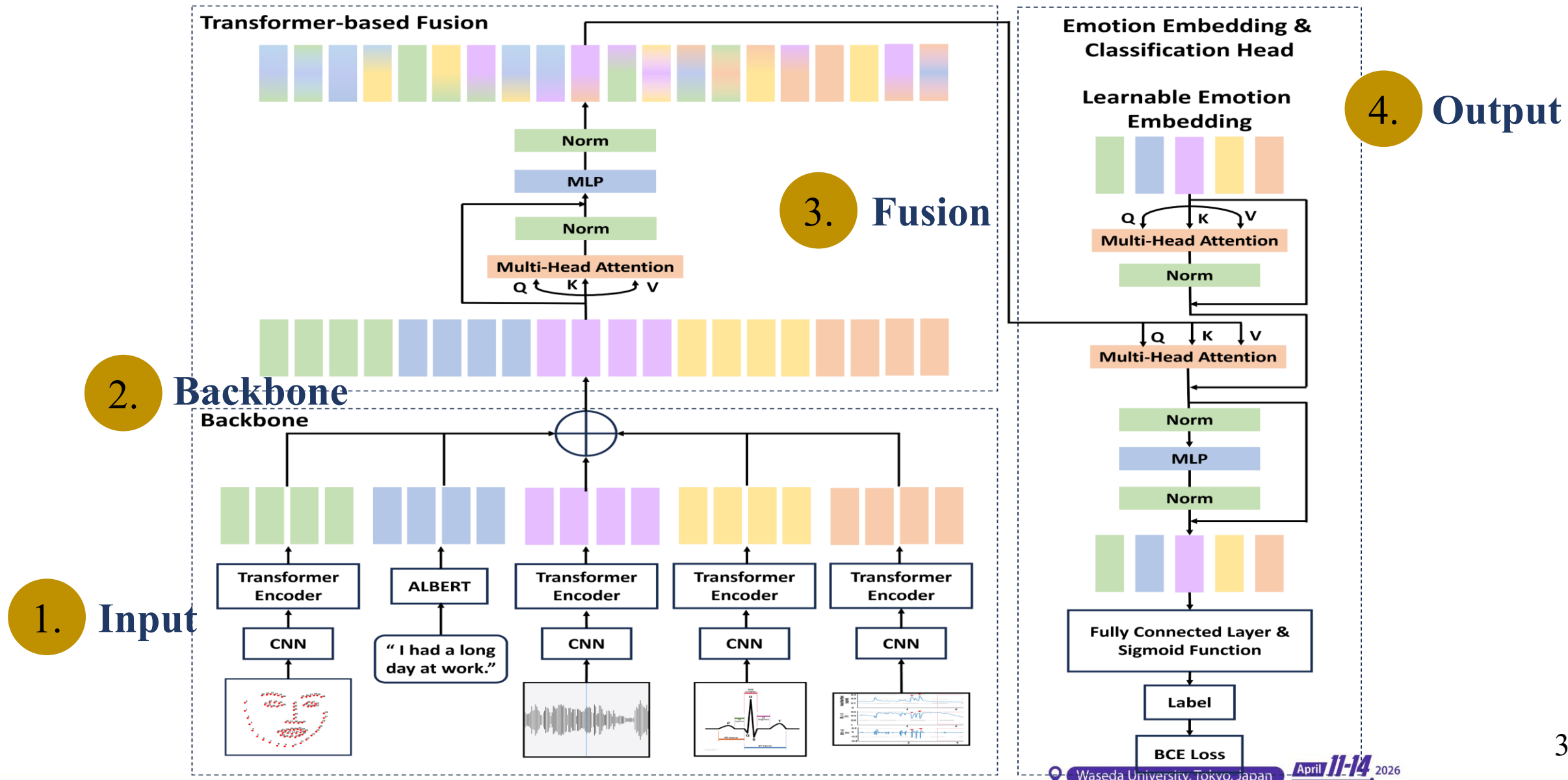


Depression

Emotion Recognition is a Big Data Challenge

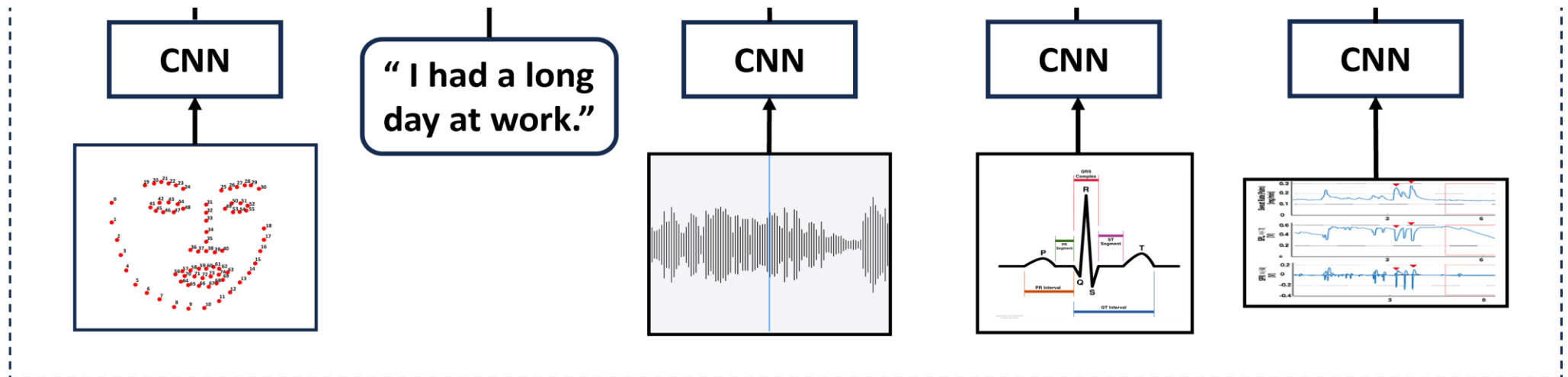


The Proposed Framework: From Signal to Sentiment



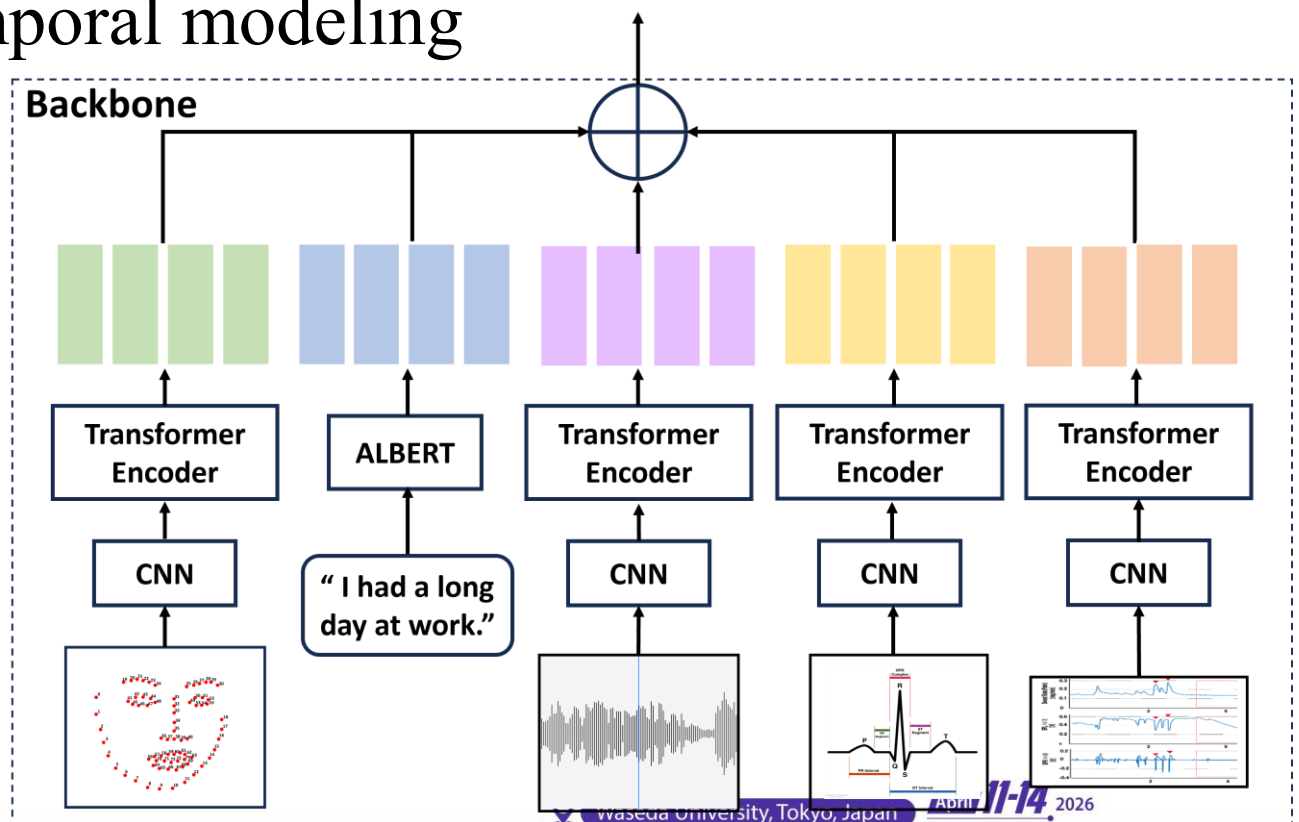
Heterogeneous Multimodal Inputs

- Four heterogeneous modalities
- Behavioral, semantic, and physiological signals
- Different sampling rates and dimensions
- High-variety big data characteristics



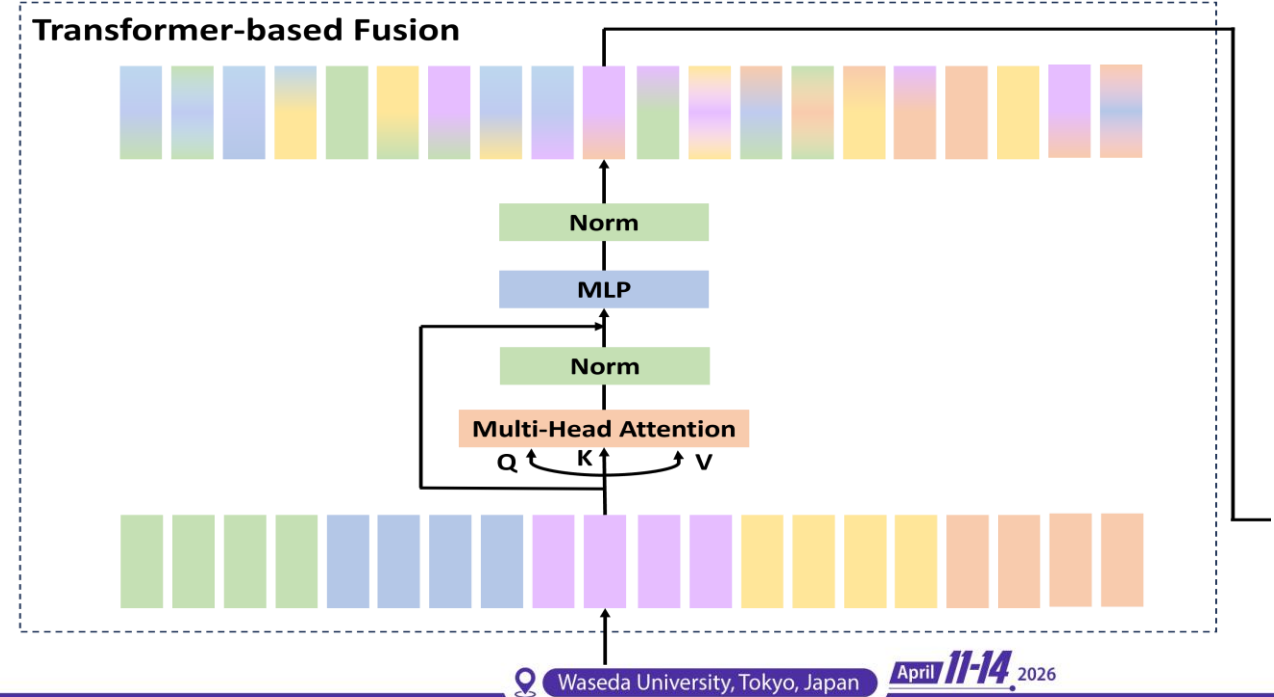
Modality-Specific Feature Extraction

- Dedicated encoder per modality
- CNN for spatial/physiological signals
- ALBERT for semantic representation
- Transformer encoder for temporal modeling
- Parallelizable architecture



Cross-Modal Transformer Fusion

- Embedding alignment into shared space
- Multi-head self-attention
- Cross-modal interaction modeling
- Global dependency learning



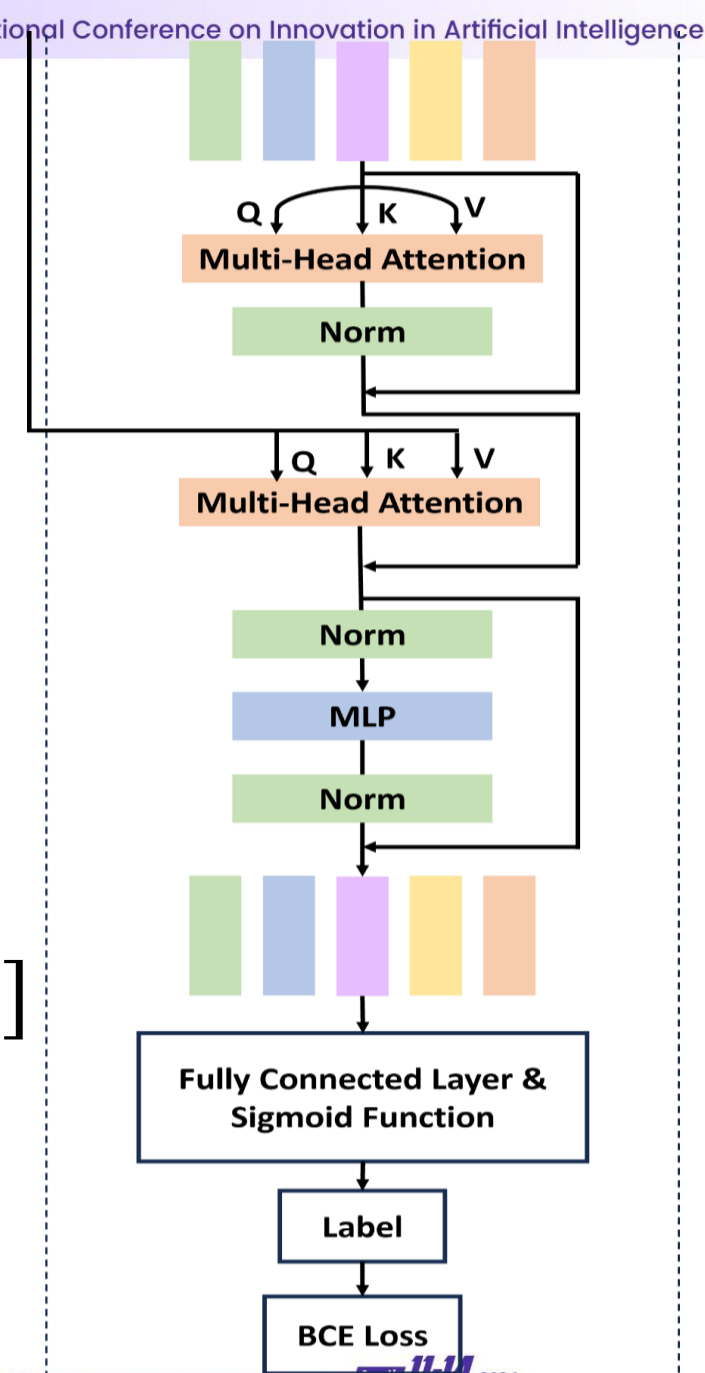
Embedding and Classification

- Learning emotion embeddings
- Emotion-specific attention
- Fully connected classifier
- Weighted BCE for class imbalance

$$h_e = \text{Attention}(q_e, Z, Z)$$

$$\mathcal{L} = - \sum_e w_e [y_e \log \hat{y}_e + (1 - y_e) \log(1 - \hat{y}_e)]$$

4. Output



Input Diversity: Capturing the Spectrum of Emotion

Visual (Face)



Captures behavioral cues via 68-point landmarks.

Text (Language)



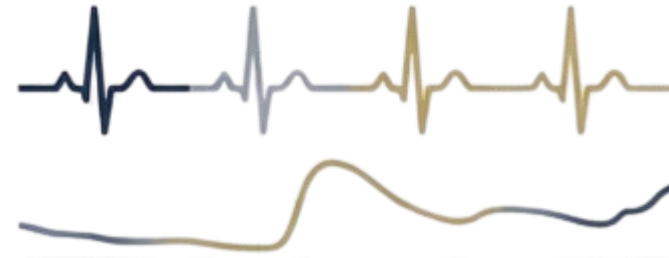
Provides semantic intent and context.

Audio (Voice)



Conveys tone, rhythm, and prosody.

Physiological (Signal)



Reveals subconscious arousal (ECG & GSR).

Feature Extraction: Behavioral Modalities



Input: 2D Facial Landmarks



Visual Processing



Vector

Output: Spatial dependency embedding

“Amazing I would like to know it more”



Visual Processing



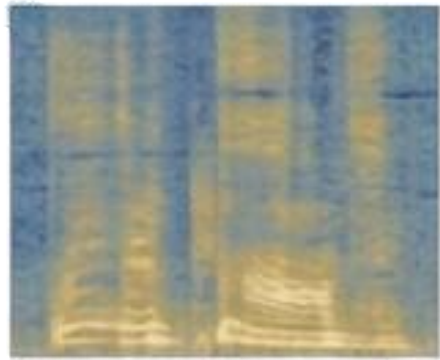
Vector

Output: Contextual [CLS] token representation

Input: Tokenized utterances

Feature Extraction: Signal Modalities

Audio Processing

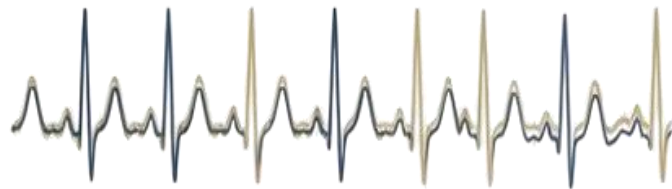


Vector

Input: Mel-spectrograms / MFCCs

Output: Acoustic embedding

Physiological Processing

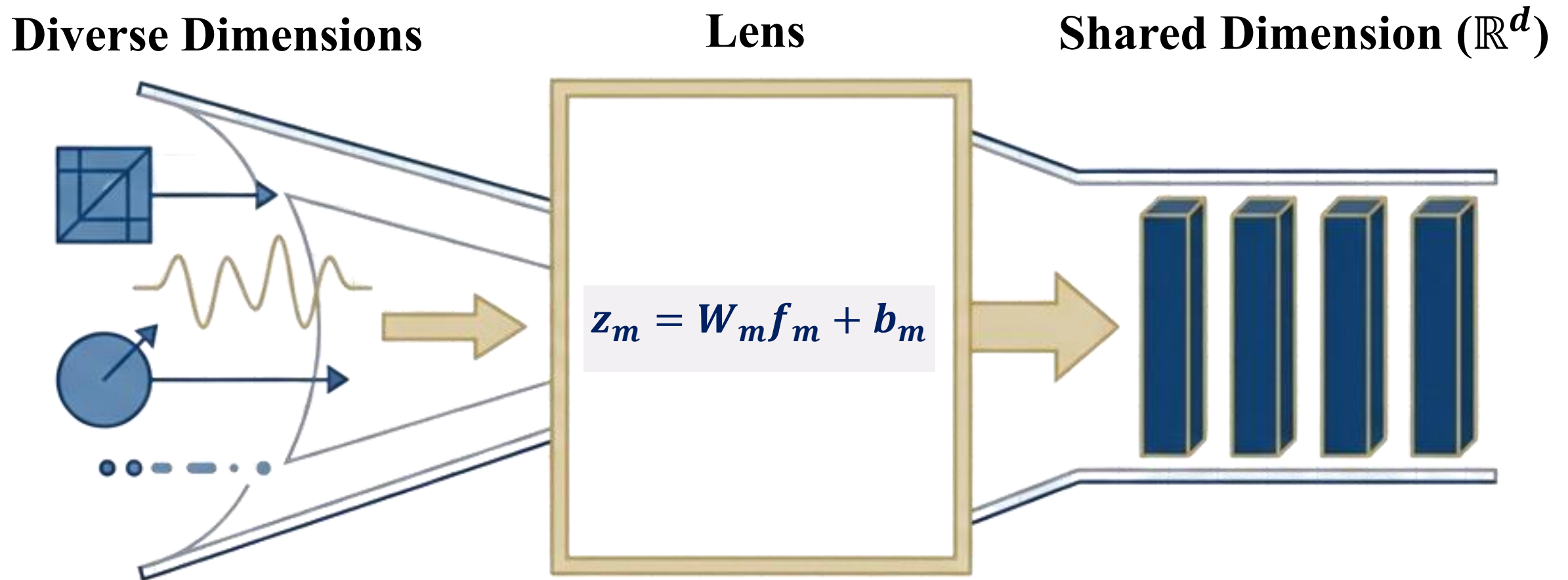


Vector

Input: Temporal time-series

Output: Physiological embedding

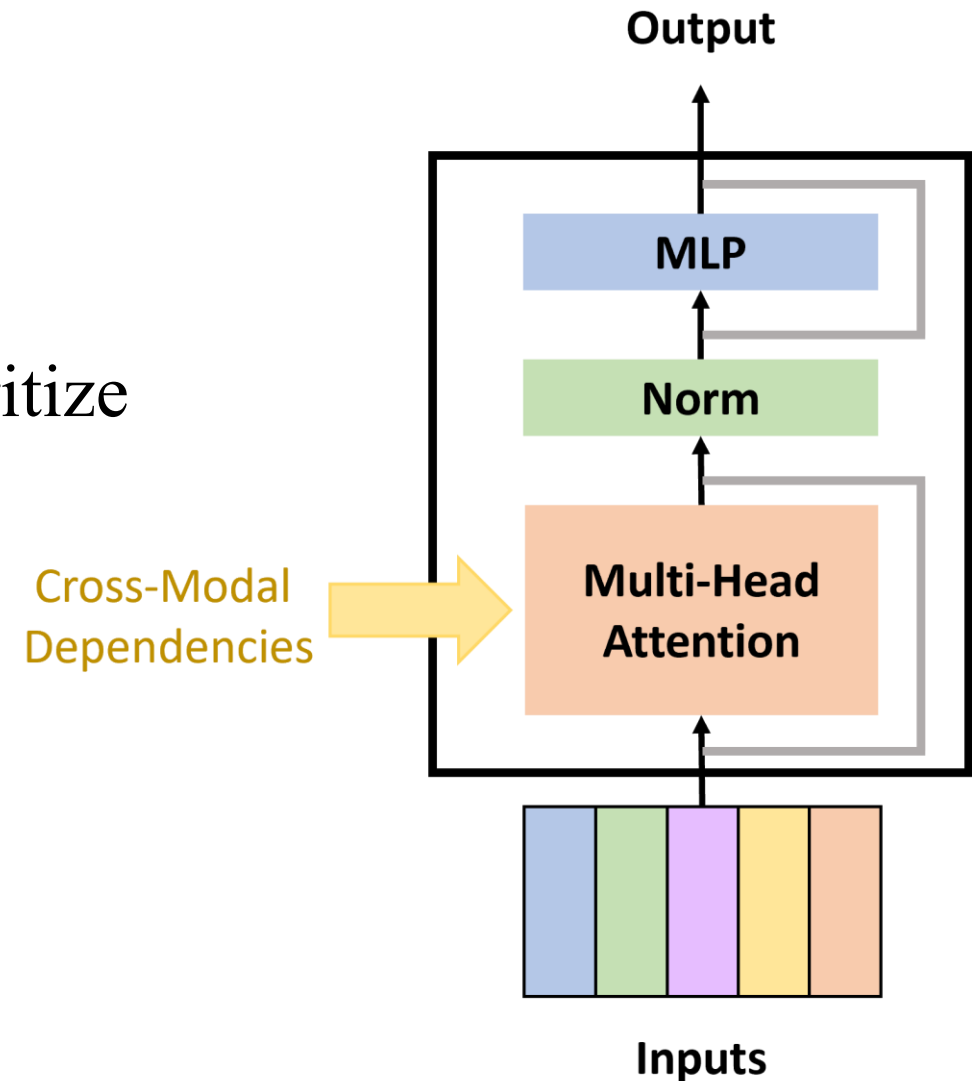
Creating a Shared Latent Space



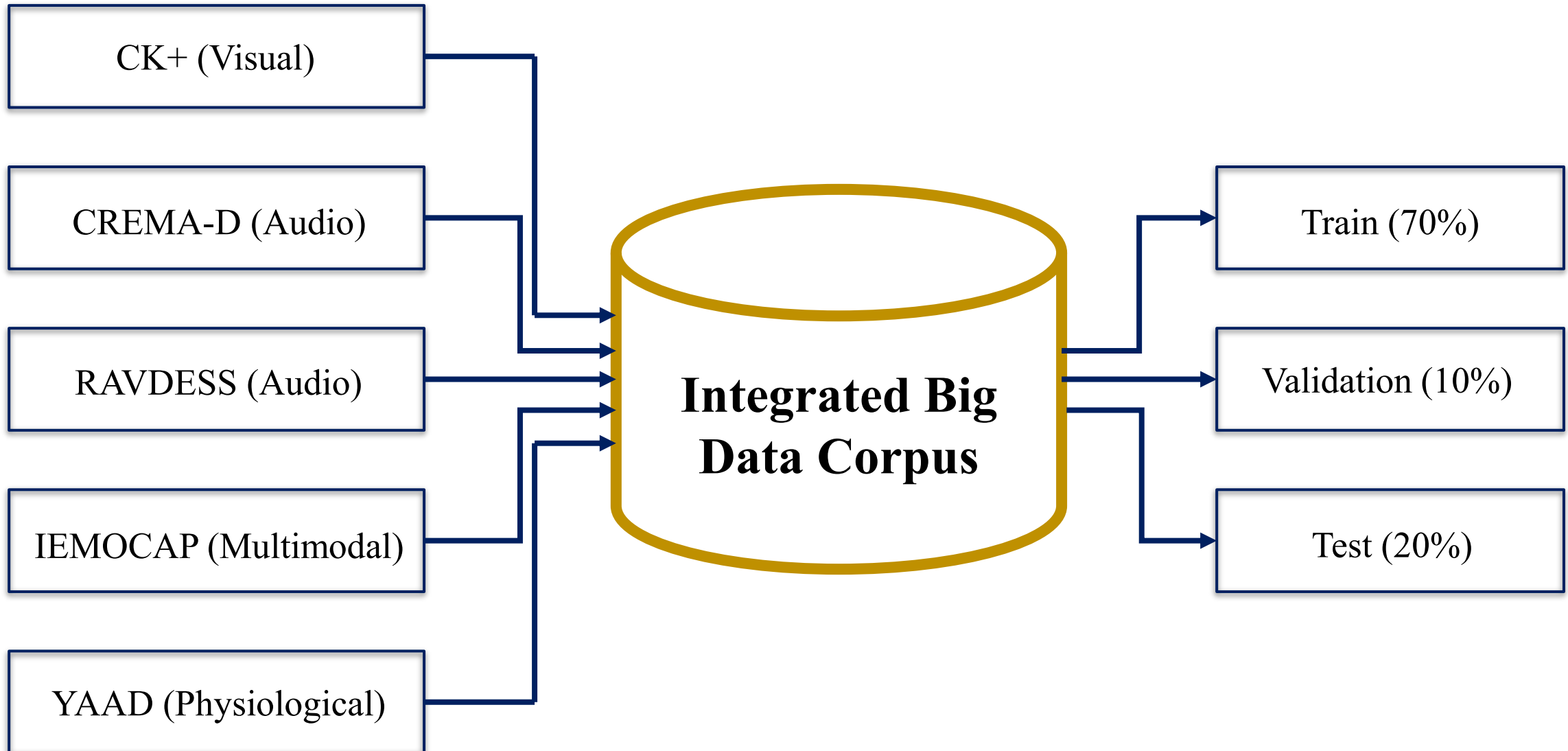
Modality-specific projection matrices align all features for efficient fusion.

Transformer-Based Fusion Mechanism

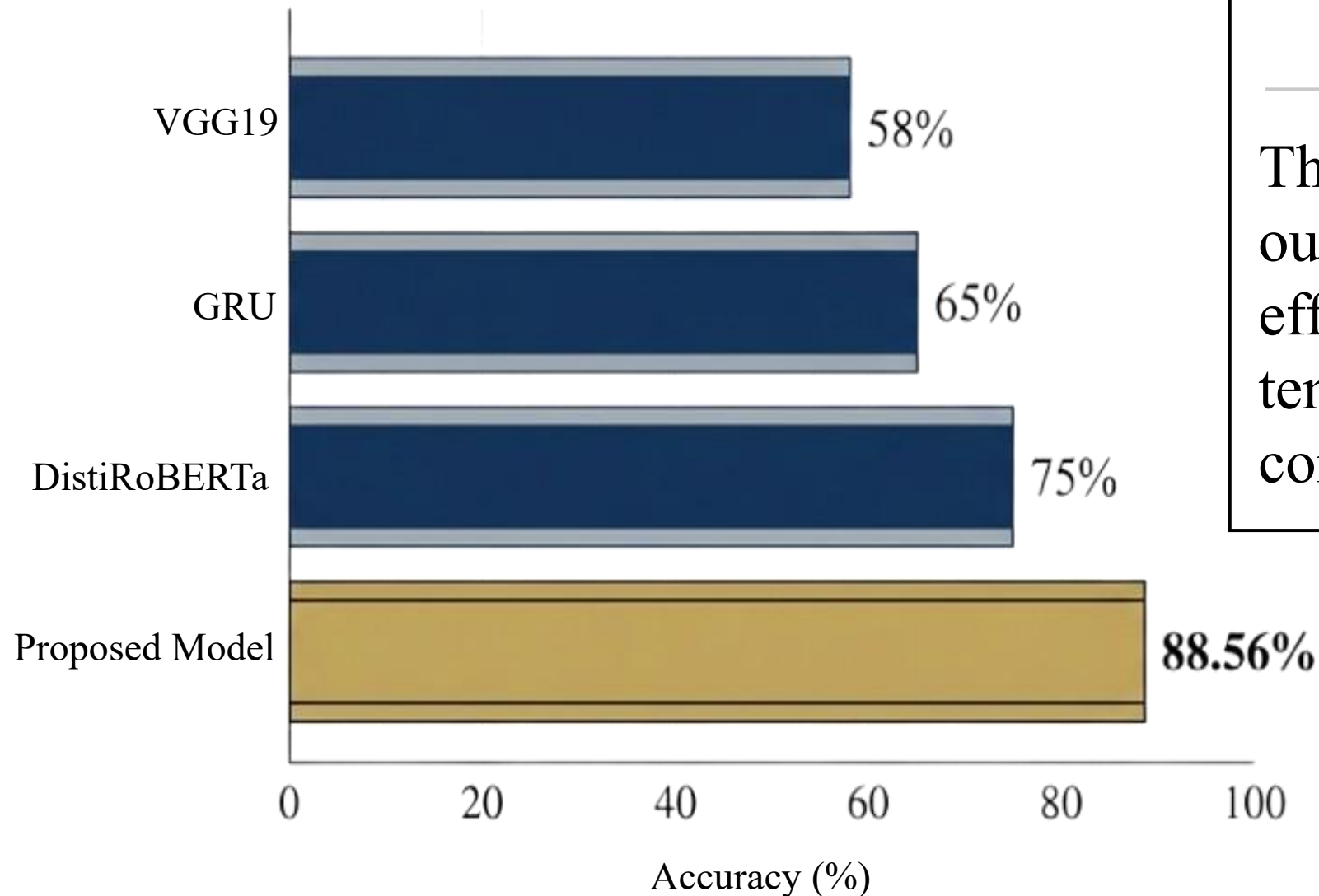
- 4 Self-Attention Layers
- 4 Attention Heads per layer
- Hidden dimension: 256
- Dynamic Weighting: Learns to prioritize relevant signals.



Experimental Setup: Integrated Big Data Corpus



Benchmarking Performance (IEMOCAP)



Result:

The Proposed Framework outperforms baselines by effectively capturing spatial, temporal, and semantic context simultaneously.

Human-Centric Validation (MOS)

Evaluated by Clinical Psychologists and Emergency Care Nurses

**Score for High-Stress
Professional Samples**

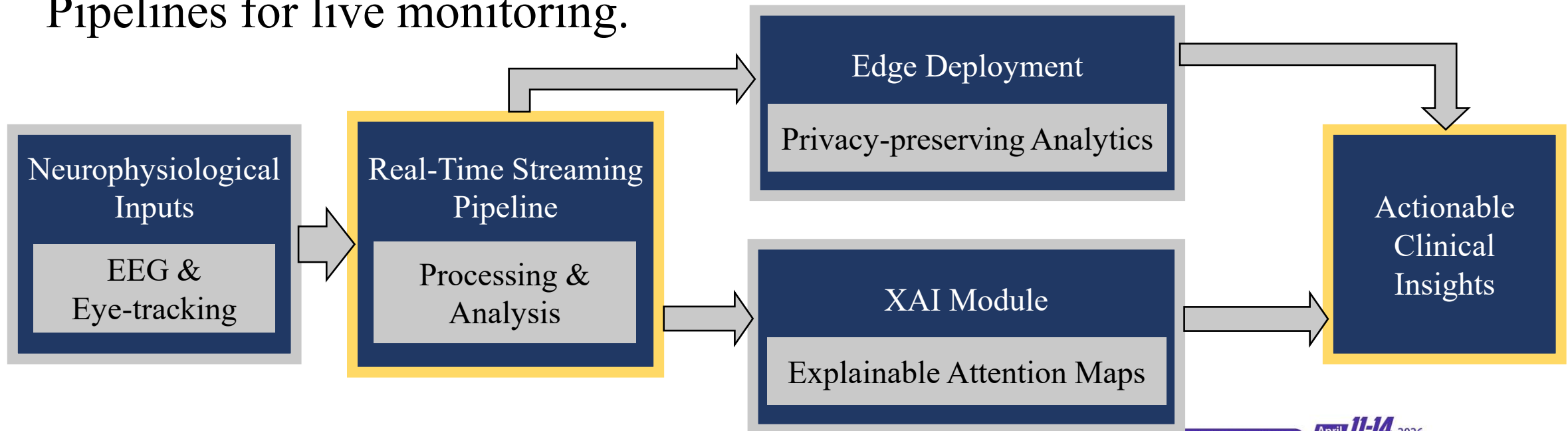


Conclusion: High perceptual alignment with expert judgment makes the system interpretable and reliable.

Future Directions:

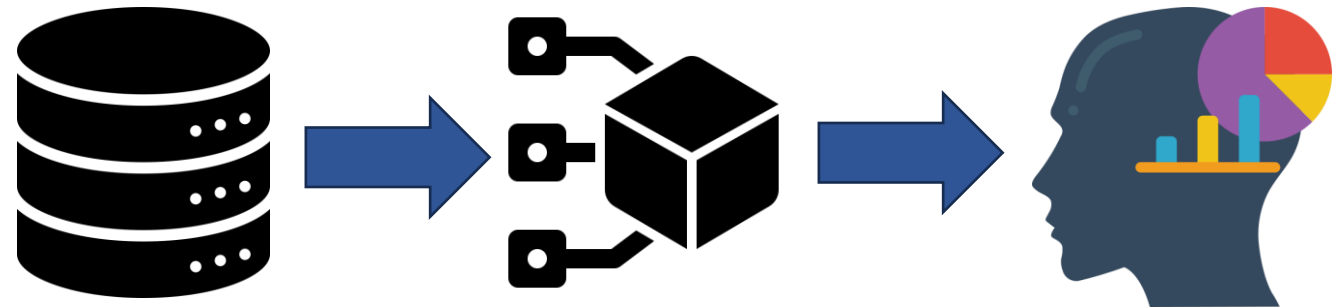
Scalability & Real-Time Deployment

- Integration of EEG (Neurophysiological cues) & Eye-tracking.
- Transition to Real-time Streaming Pipelines for live monitoring.
- Edge-based deployment for privacy-preserving analytics.
- Explainable AI (XAI) maps to visualize attention layers for clinicians.



Conclusion: Redefining Mental Health Analytics

- Scalable multimodal architecture
- Cross-modal embedding alignment
- Transformer-based heterogeneous fusion
- Robust emotion recognition across datasets



Why Multimodal Instead of Unimodal

- Emotions are expressed through multiple modalities, so combining them improves robustness and captures more complete information.

Why High-Stress Environments

- High-stress environments are more likely to cause burnout and depression, making emotion monitoring especially important.

Why Transformer Architecture

- **Transformers can capture long-range dependencies and model interactions across different modalities.**

Multimodal Fusion Strategy

- **We align features in a shared space and use attention to learn relationships across modalities.**

Modalities Used in the Framework

- We use facial features, speech, text, ECG, and GSR signals.

Handling Noisy & Missing Data

- The attention mechanism dynamically adjusts the importance of each modality to reduce noise impact.

Performance vs. Baselines

- **Our model achieves higher accuracy by leveraging complementary information from multiple modalities.**

Dataset Selection Rationale

- **These datasets cover multiple modalities and are widely used benchmarks.**

Real-World Applications

- It can be used for mental health monitoring, stress detection, and intelligent healthcare systems.

Future Research Directions

- We plan to add more signals, enable real-time processing, and improve model interpretability.

Limitations of the Study

- **The datasets may not fully reflect real-world scenarios, and data collection is challenging.**

Why Not Only Physiological Signals

- **Physiological signals lack context, so combining modalities gives a more complete understanding.**

Data Split Strategy (70/10/20)

- **Training learns the model, validation tunes parameters, and testing evaluates performance.**

Source of Performance (88.56%)

- **Multimodal fusion and attention mechanisms improve robustness and reduce ambiguity.**

Mean Opinion Score (MOS 9.5)

- It reflects high agreement between the model's predictions and expert judgments.

Dataset

- Our study does not rely on a single dataset. Instead, we integrate multiple publicly available datasets, such as CK+, IEMOCAP, and YAAD, to construct a multimodal big data corpus.

Did you obtain IRB approval for this study?

- **Yes. The datasets we used were originally collected under ethical approval by their respective research teams, including informed consent from participants. In our study, we only used these publicly available and anonymized datasets for secondary analysis, and no additional human subject recruitment was involved.**